Two leaf nodes → N, S, S

Tutorial-10

S.	F1 (Age)	F2 (Income)	F3 (Experience)	Tar
S1	24	35	1	
S2	30	50	3	
S3	45	40	10	
S4	35	30	7	
S5	28	55	2	

Target Encoding

Yes = +1

No = -1

1) $w_i = \frac{1}{N} = \frac{1}{5} = 0.2$

2) Decision stump (weak learner) - Income (g

Income < 38 → No, ✗

≥ 38 → yes, ✓

N = 5

3) $\varepsilon_1 = \sum_{i=1}^{N} w_i I = [0 \times 0.2 + 0 \times 0.2 + 0 \times 0.$

4) $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$

$= \frac{1}{2} \ln\left(\frac{1}{0}\right) = \infty$

5) update weights. $= y_i \alpha_1 h_1(x_i)$

6) Final prediction

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right)$$

$$= \text{sign}(\alpha_1 h_1(x_1)) = \text{sign}(\alpha_1 h_1(x_1))$$

$$H(y) = h_1(x_1)$$

Q2:

S	Age r_1	Income r_2	Experience r_3	Job offer r_4	Tag	pre	winew	weight normalized
S1	24	35	1	No	-1	-1	0.1	0.125
S2	30	50	3	Yes	+1	+1	0.1	0.125
S3	45	40	10	Yes	+1	+1	0.1	0.125
S4	35	30	7	No	-1	-1	0.1	0.125
S5	28	55	2	No	-1	+1	0.4	0.5

1) $w_i = 1/N = 1/5 = 0.2$

2) Decision stump (weak learner) - Income (given in ques)

Income $< 38 \rightarrow -1$

$\geq 38 \rightarrow +1$

3) $\varepsilon_t = \sum_{i=1}^{N} w_i \cdot I$

$= [0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 1]$

$= 0.2$

4) $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$

$= \frac{1}{2} \ln\left(\frac{1-0.2}{0.2}\right) = \frac{1}{2} \log\left(\frac{0.8}{0.2}\right) = 0.693$

5) update weight

$y_1 = w_i^{old} e^{-y\alpha_1 h_1(x_1)}$

$= 0.2 e^{-(-1)(0.69)(-1)}$

$= 0.2 e^{-(0.69)} = 0.100$

$y_2 = 0.2 e^{-(1)(0.69)(1)} = 0.2 e^{-0.69} = 0.100$

$y_3 = 0.2 e^{-(1)(0.69)(1)} = 0.100$

$y_4 = 0.2 e^{-(-1)(0.69)(-1)} = 0.100$

$y_5 = 0.2 e^{-(-1)0.69(1)} = 0.2 e^{0.69} = 0.398$

$i=1$

$w_2 = \frac{0.1}{0.8} = 0.125$, $w_3 = w_4 =$

) final prediction 8) $h_1(x_2) = +1$

a) $H(x) = \text{sign}(\alpha_1 h_1 + \alpha_2 h_2 + \alpha_3 h_3 + \ldots)$

$= \text{sign}(0.693 \times 1$

- if both misclassifiers are have same
sign in 8th step we tak
that sign, if both missclas
are opposite (+1, −1) then, 8th
is taken from Normalized weights,
which have highest weigh
value in αw.